"Fatty, Fatty, Two by Six?: Family Income and Child Obesity"

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Question and Empirical Framework

1. **Research question:**
   1) How does family income affect child obesity?
   2) Why does such relationship occur?

2. **Empirical Framework**

$$y_{it} = \alpha f(\textit{family income}_{it}) + X_{it}\beta + [\nu_i] + \epsilon_{it}$$

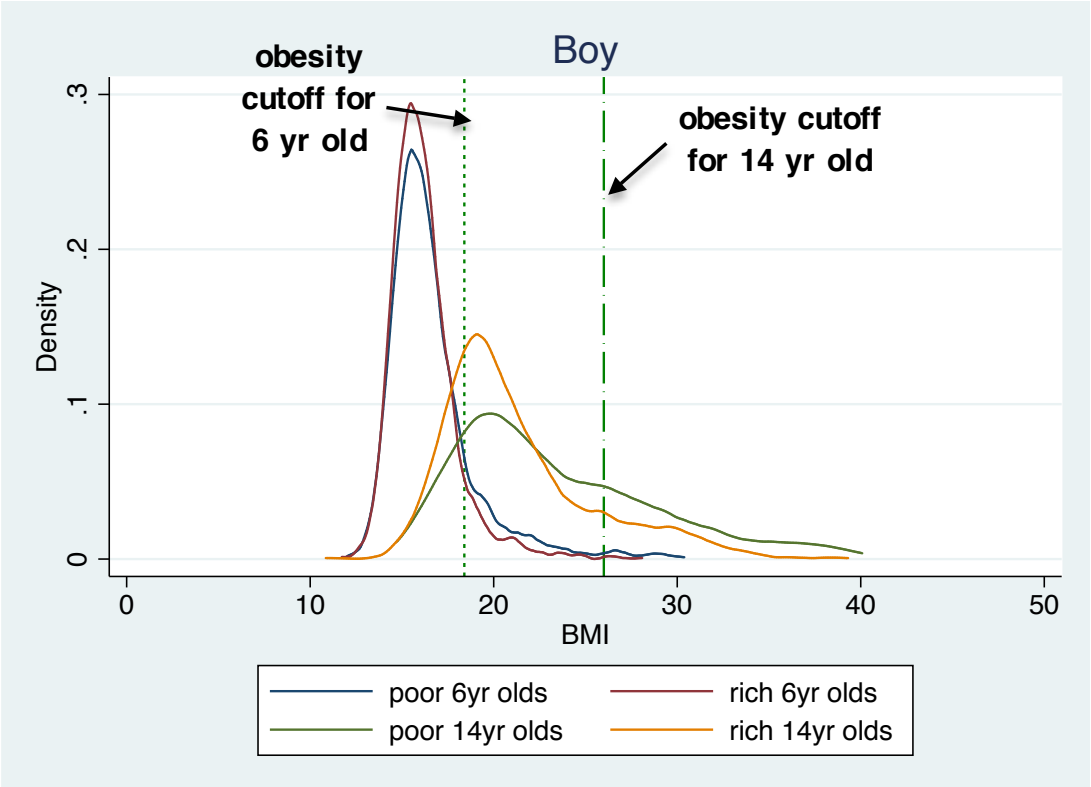
y_{it} obesity status or BMI

X_{it} vector of individual characteristics

ν_i child FE

1. Increasing Obesity Gap

Figure 3: BMI distribution by age and income, ECLS-K



- As children age, BMI distribution shifts to the right as the variance increases.
- Low-income children's BMI distribution expands more with a thicker right tail.

2. Increasing Income Gradient

- **Negative Relationship**: Income and obesity rate are negatively related
- **Increasing Gradient**: as children age, the relationship between income and child obesity becomes stronger.
- **Non-linearity**: Quantity versus Quality

3. Source of weight gain

Low-income children are gaining weight faster, why?

- **Genetics or other unobserved individual heterogeneities**

  - Inclusion of child FE doesn't eliminate increasing gradient

- **Higher calorie consumption**

  - Income Constraint: high calorie food is cheaper

  - Lack of Alternatives: they live in a "food desert" with lack of healthy food options

- **Lower calorie expansion**

  - Lack of Facilities: playgrounds or gyms are not readily available

  - Safety Concern: Low-income neighborhoods may not be safe for children to play outside

  - Maternal employment: mothers lack resources to monitor children

4. EITC as a natural experiment

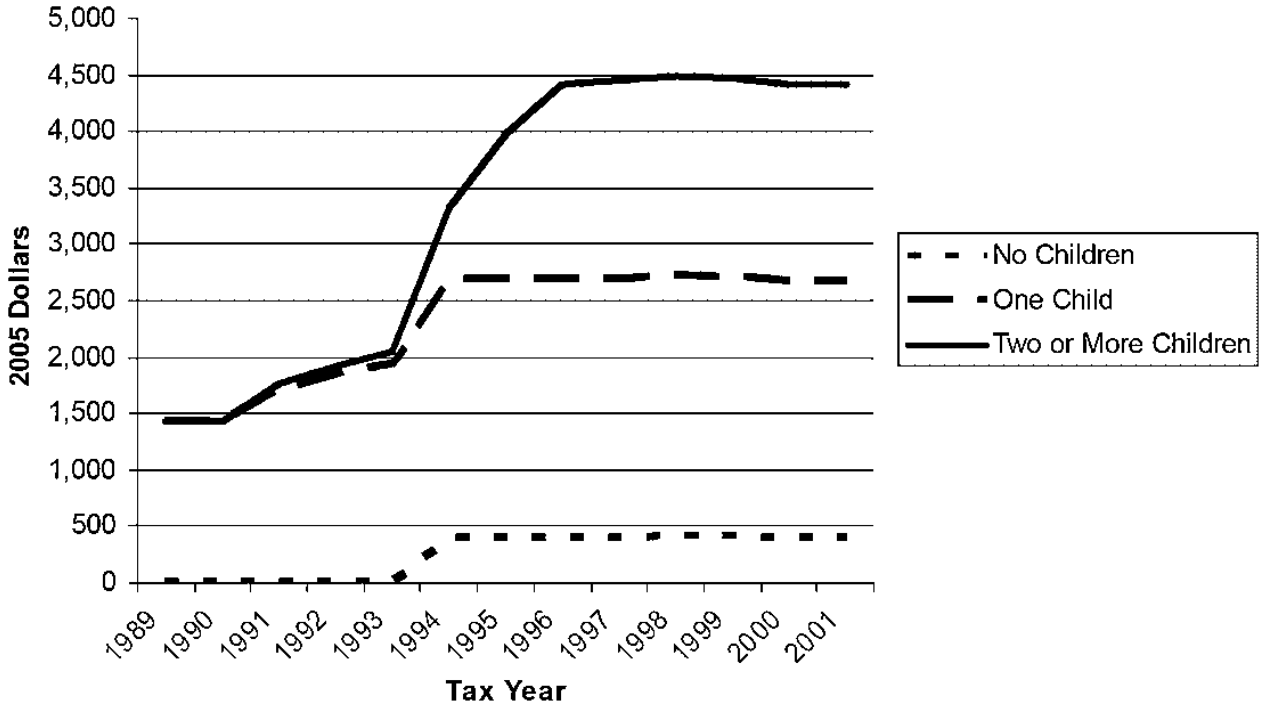


Figure 2. Maximum value of Federal EITC benefits ($ 2005), by eligible children

1. Exploit exogenous variation in income resulting from expansion of the Earned Income Tax Credit (EITC)
   - Use maximum value of EITC as an instrument variable for TSLS
   - Measure the effect of income on nutrition